

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year **Backlog Examination**, 2018
Department of Materials Science and Engineering

Ch 1227
Organic Chemistry

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What types of hybridizations are found in alkanes, alkenes and alkynes? Explain the differences including stereochemical features? (10)
- (b) How would you account for the fact that CH₄ and CCl₄ have no dipole moment, whereas CHCl₃ has a dipole moment? (08)
- (c) What is hydrogen bonding? How does it modify the properties of organic compounds? (10)
- (d) Define bond order. Briefly discuss the relationship between bond length and bond order. (07)
- Q 2. (a) Discuss the mechanism of ionic polymerization. (12)
- (b) What is conducting polymer? Write down the advantages of conducting polymer over conventional polymer. (11)
- (c) Distinguish the following: (12)
- (i) Natural polymers and synthetic polymers
 - (ii) Organic polymers and inorganic polymers
 - (iii) Homopolymers and copolymers
- Q 3. (a) Define aromaticity and explain Huckel's rule. Will cyclooctatetraene show aromatic character? (12)
- (b) Phenols are acidic in character, while alcohols are almost neutral. Explain. (10)
- (c) Explain the mechanism of electrophilic substitution in benzene. (13)
- Q 4. (a) Define isotactic, syndiotactic and atactic polymer with examples. (09)
- (b) How are aromatic amines defined? Describe suitable reactions by which different amines can be distinguished from each other (10)
- (c) Pyrrole is weakly acidic whereas Pyridine is weakly basic – Explain. (06)
- (d) Why dimethylamine has a higher boiling point than trimethylamine even though the latter has an approximately higher molecular weight? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) What are the different types of electronic transitions possible in a molecule during UV spectroscopy? Give the energy diagram of those transitions. (C)
- (b) Define the term Chromophore and Auxochrome. Give examples. (1)
- (c) Write down the basic principle of IR spectroscopy. (1)
- Q 6. (a) S_N2 reaction proceeds with complete stereochemical inversion-Explain with example. (C)
- (b) Why an S_N1 reaction proceeds with racemization? (C)
- (c) What is Hoffman and Zaitsev's rule? Explain with example (C)
- Q 7. (a) Define N – terminal residue of polypeptides. Discuss any one method for identification of N – terminal residue of peptide. (C)
- (b) What are essential amino acids? Why α -amino acids (except glycine) are optically active? (C)
- (c) What is meant by denaturation of Proteins? What happens when amino acids are reacted with ninhydrin? (C)
- (d) Write down the reason for amphoteric nature of amino acids. (C)
- Q 8. (a) What happen when cellulose react with concentrated nitric acid in presence of sulphuric acid? Give reaction. (C)
- (b) A conjugated diene absorbs at higher wavelength compared to diene in which double bonds are isolated. Explain with example. (C)
- (c) 3° carbocation is more stable than 1° Carbocation. Why? (C)
- (d) Why is methanol a good solvent for UV but not for IR determination? (C)